Romy Mondello

AI & FULL-STACK ENGINEER · AGENT INFRASTRUCTURE

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SUMMARY

I build agent systems and the infrastructure they run on. Recently that's meant a merged contribution to qq, an open-source Rust coding agent, and a counterexample on arXiv whose proof I verified in Lean 4. Before that I co-designed the agent architecture at Further and wrote backend services at Knox Financial.

EXPERIENCE

Further (withfurther.com)

Feb 2025 – Apr 2025

Research Engineer (Contract)

- Co-designed the LangGraph agent and data pipeline architecture with the CTO, then built the pipelines that pull Zillow and mortgage API data through retrieval and rule-based risk checks.
- Got the team using Cursor and coding agents day to day, and fixed LaTeX rendering and API timeouts in the AI Genius web app.

Pillzumi

Dec 2024 – Feb 2025

Machine Learning Engineer (Contract)

- Built long-running autonomous X agents on elizaOS that answered DMs and remembered past conversations without tripping API rate limits.

Verizon Business

Apr 2024 – Dec 2024

Customer Solutions

- Managed enterprise accounts, scoping software and network solutions with customers' technical teams.

Verus AI

2022 – 2023

Data Scientist

- Built GCP pipelines that moved on-chain data into BigQuery and Vertex AI.
- Trained recurrent models to forecast spend, usage, and customer growth, and kept an eye on data quality once they were in production.

Knox Financial

Jul 2020 – Mar 2021

Backend Software Engineer

Real-estate investment platform, Boston.

- Built the housing-search backend on real-estate data feeds and AWS and Google APIs.
- Wrote data ingestion pipelines, Postgres-backed forecasting services, and the REST APIs the web app used.
- Fixed flaky tests, added pre-deploy checks to CI, and reviewed backend PRs.

OPEN SOURCE & RESEARCH (MAY 2025 – PRESENT)

Published a 10-page arXiv preprint giving a counterexample for separable Jacobian maps in characteristic two, with the full proof machine-checked in Lean 4 and Mathlib.
qq, an open-source coding agent in Rust: merged PR #72 adding review checkpoints and reasoning-effort settings that work across model providers.
Agent Cloud (prototype, with a cofounder): Temporal runs long agent jobs while Postgres tracks each tenant's budgets and cancellations, so a job resumes after a crash.
Ferrite, a Rust-first web framework prototype with server-side rendering, file-based routing, and hydration through WASM.
Wrote a 166,700-neuron fruit-fly connectome simulator in MLX on Apple silicon.

EDUCATION

Associate of Science, Computer Science, Bunker Hill Community College

SKILLS

Languages: TypeScript, Python, Rust, Lean 4
Backend and infra: Node.js, FastAPI, Axum, PostgreSQL, Temporal, Redis, Docker, Cloudflare Workers, AWS, GCP, BigQuery
Frontend: React, Next.js, Tailwind
AI and ML: LangGraph, RAG, elizaOS, PyTorch, MLX